

 Part 1 – Question Paper**10-Mark Questions**

1. Explain the fundamental design principles that ensure the security and efficiency of a block cipher.
2. Provide a comprehensive analysis of the Data Encryption Standard (DES) algorithm, including its structure and round operations.
3. Discuss the Advanced Encryption Standard (AES) architecture, detailing its round-based processing and key length variations.
4. Examine the internal transformation functions used in each round of AES encryption.
5. Compare the Data Encryption Standard (DES) and Advanced Encryption Standard (AES) in terms of security, structure, and performance.

5-Mark Questions

6. Describe the concepts of Confusion and Diffusion and their roles in symmetric ciphers.
7. Explain the Avalanche Effect and why it is a critical requirement for encryption algorithms.
8. Outline the process of Key Schedule and round key generation in the DES algorithm.
9. Discuss the parameters and rounds required for different versions of AES (AES-128, 192, and 256).
10. Explain the decryption process in AES and how it relates to the encryption stages.
11. Describe the role and design of the Substitution Box (S-box) in providing non-linearity.
12. Detail the structure of the Symmetric Cipher Model, including the roles of the sender and receiver.
13. Analyze the security vulnerabilities of DES that led to the adoption of AES.

2-Mark Questions

14. Define Symmetric Encryption and its alternative name.
15. What is the difference between Plaintext and Ciphertext?
16. Define the term "Block Cipher."
17. State the input, output, and key sizes used in the DES algorithm.
18. What is the purpose of the Initial Permutation (IP) in DES?
19. Define "Key Schedule" in the context of symmetric ciphers.
20. What block size is used by the AES algorithm?
21. Define the "Add Round Key" step in AES.
22. What is "Enciphering" and "Deciphering"?
23. Mention the NIST adoption years for both DES and AES.

EXAM POV NOTES (Q&A FORMAT)**✦ Part 2 – Complete Answer Bank****10-Mark Questions**

Q1. Explain the fundamental design principles that ensure the security and efficiency of a block cipher.

Introduction: A block cipher is an encryption algorithm that processes data in fixed-size segments, usually 64 or 128 bits, utilizing complex mathematical principles to ensure data protection.

- **Number of Rounds:** This principle defines the iterative nature of the algorithm, where multiple rounds increase the complexity of the encryption process to make it more secure against analysis.
- **Design of Function F:** The Round Function is the core component of a block cipher's structure, and its internal complexity directly determines how difficult it is for an attacker to perform cryptanalysis.
- **Confusion Property:** This property ensures the ciphertext is a complex function of the key and plaintext, which hides the relationship between them and prevents attackers from easily guessing the secret key.
- **Diffusion Property:** Diffusion ensures that even a small change in the plaintext causes a significant and widespread change across the ciphertext, making it difficult to analyze encryption patterns.
- **Key Size Selection:** A block cipher must use a sufficiently large key, such as 128 bits, to provide a massive pool of possible keys that effectively prevents brute-force guessing attacks.
- **Key Schedule Design:** This algorithm must generate round keys that are independent and unpredictable to resist attacks that exploit weak keys or specific mathematical properties of the cipher.
- **Block Size Choice:** The block size must be large enough to prevent attackers from finding and exploiting statistical patterns that might exist in the original plaintext data.
- **Non-linearity (S-boxes):** Designers use non-linear Substitution-boxes to provide confusion, as linear structures are highly vulnerable to mathematical attacks that exploit predictable properties.
- **Avalanche Effect:** This requirement ensures that any change in input (plaintext or key) leads to a complete transformation of the output, ensuring no bit-by-bit relationship remains visible.
- **Security Analysis:** Every design must undergo rigorous testing against known methods like differential or linear cryptanalysis and resistance to physical implementation threats like side-channel attacks.

Q2. Provide a comprehensive analysis of the Data Encryption Standard (DES) algorithm, including its structure and round operations.

Introduction: DES is a symmetric block cipher adopted by NIST in 1977 that processes 64-bit blocks of data through 16 rounds of transformation using a Feistel structure.

- **Input and Output Size:** The algorithm is designed to take a 64-bit plaintext block as input and produce a corresponding 64-bit ciphertext block as the final output.
- **Main Key Transformation:** While the initial input key is 64 bits, it is immediately converted into a 56-bit effective subkey by removing parity bits before further processing.
- **Initial Permutation (IP):** Before the main rounds begin, the plaintext block undergoes a bit-shuffling process that rearranges the data according to a fixed permutation table.
- **Round Structure:** The core of DES consists of 16 rounds of iterative processing, where each round applies a unique 48-bit key to transform the data.
- **Expansion Permutation:** During each round, the 32-bit right half of the data is expanded to 48 bits so it can be combined with the 48-bit round key.
- **Round Key Generation:** A Key Schedule generates a unique 48-bit key for every round using circular shifts and permutations on the 56-bit subkey.
- **S-box Substitution:** The output of the key-XOR operation is passed through eight S-boxes, which provide critical non-linearity by substituting 6-bit inputs for 4-bit outputs.
- **P-box Permutation:** Following substitution, a permutation box shuffles the 32 bits of data to ensure that the influence of individual bits is spread across the block.
- **32-bit Swap:** After the 16th round is completed, the algorithm performs a final swap of the left and right 32-bit data halves before the final stage.
- **Inverse Initial Permutation:** The last step is the Inverse IP, which maps the bits back to their final positions to yield the complete 64-bit ciphertext.

Q5. Compare the Data Encryption Standard (DES) and Advanced Encryption Standard (AES) in terms of security, structure, and performance.

Introduction: DES and AES represent different eras of cryptography, with AES being the modern replacement for the older, less secure DES algorithm.

Feature	Data Encryption Standard (DES)	Advanced Encryption Standard (AES)
NIST Adoption	Adopted in 1977.	Adopted in 2001.
Block Size	Processes 64-bit blocks.	Processes 128-bit blocks.
Effective Key Size	Uses a 56-bit subkey.	Uses 128, 192, or 256 bits.
Rounds	Fixed at 16 rounds.	Variable: 10, 12, or 14 rounds.
Cipher Structure	Built on the Feistel structure.	Based on Substitution-Permutation Network.
Data Processing	Operates primarily on bits.	Operates on bytes of data.
Security Status	Vulnerable due to small key size.	Highly secure and widely used today.
Implementation	Efficient in older hardware.	Harder to implement but highly efficient.
Transformation	Uses expansion and P-boxes.	Uses MixColumns and ShiftRows.
Round Keys	Uses 48-bit round keys.	Round keys match the 128-bit block.

5-Mark Questions

Q6. Describe the concepts of Confusion and Diffusion and their roles in symmetric ciphers.

Introduction: Confusion and diffusion are the primary properties identified to protect data by making the relationship between the key, plaintext, and ciphertext difficult to analyze.

- **Confusion Definition:** Confusion ensures that the ciphertext is a complex mathematical function of both the secret key and the plaintext message.
- **Confusion Purpose:** Its specific role is to frustrate attackers by making it nearly impossible to deduce the key even if the output ciphertext is known.
- **Diffusion Definition:** Diffusion ensures that a small change in the plaintext results in a significant and widespread change in the resulting ciphertext.
- **Diffusion Purpose:** This property spreads the characteristics of individual plaintext bits throughout the ciphertext to hide patterns and statistical structures.
- **Implementation Mechanism:** Symmetric ciphers achieve these properties through a series of linked operations that involve replacing (substitution) and shuffling (permutation) data.

Q13. Analyze the security vulnerabilities of DES that led to the adoption of AES.

Introduction: While DES was once a global standard, it was eventually replaced by AES because its original design could no longer withstand modern computing power.

- **Short Key Length:** The primary flaw of DES is its 56-bit key size, which provides only 2^{56} possible combinations for an attacker to test.
- **Brute-Force Success:** Because of the limited key space, modern computers can perform exhaustive searches to find the correct key in a very short amount of time.
- **Security Vulnerabilities:** NIST officially retired DES after identifying various security vulnerabilities that made it untrustworthy for protecting sensitive modern data.
- **Hardware-Specific Design:** DES was originally designed for older hardware efficiency, making it slower and less secure in software-driven modern environments.
- **Superior AES Security:** AES was adopted because it is much stronger than DES, offering larger key lengths and a structure that is more resistant to cyber threats.

Section C: 2-Mark Questions

Q14. Define Symmetric Encryption and its alternative name.

- **Basic Definition:** It is an encryption method that uses exactly one secret key for both the encryption of plaintext and the decryption of data.
- **Alternative Name:** Because only one key is shared for the entire process, it is also known as Single-Key Encryption.

Q15. What is the difference between Plaintext and Ciphertext?

- **Plaintext:** This is the original, readable message that is intended to be communicated between a sender and a receiver.
- **Ciphertext:** This is the encoded, unreadable format of the original message produced after the encryption process is applied.

Q17. State the input, output, and key sizes used in the DES algorithm.

- **Data Dimensions:** DES processes data in 64-bit blocks, producing a 64-bit output for every 64-bit block of plaintext input.
- **Key Dimensions:** The algorithm takes a 64-bit main key as input but converts it into a 56-bit subkey for the encryption process.

Q20. What block size is used by the AES algorithm?

- **Standard Size:** AES processes data in fixed blocks of 128 bits each, regardless of the key length being used.
- **Byte Equivalent:** Since AES operates on bytes, this 128-bit block size corresponds to 16 bytes of input data processed at a time.

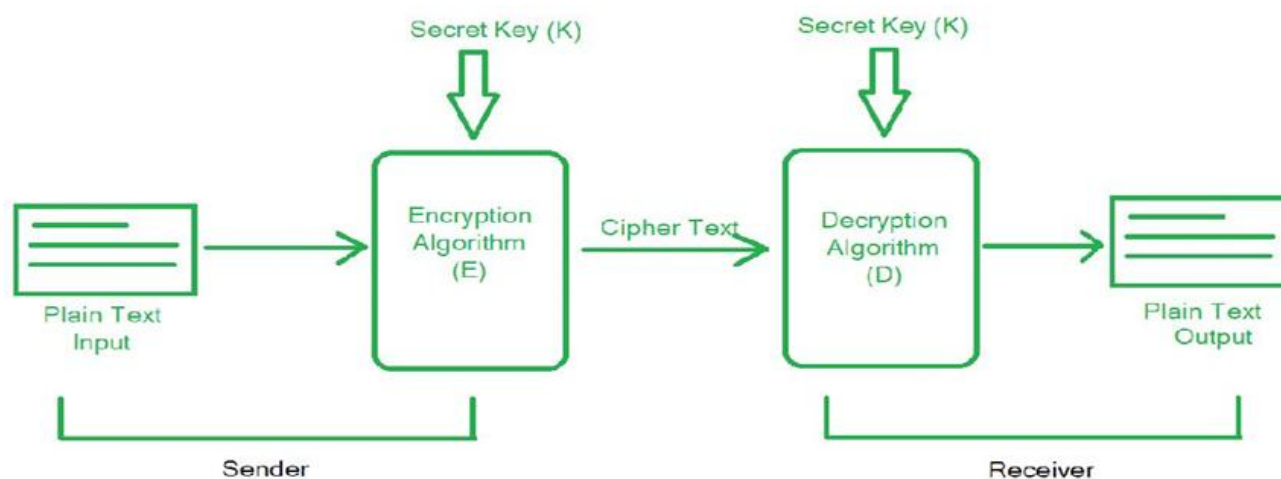
Q23. Mention the NIST adoption years for both DES and AES.

- **DES Adoption:** The Data Encryption Standard (DES) was officially adopted by NIST in 1977.
- **AES Adoption:** The Advanced Encryption Standard (AES) was officially adopted by NIST to replace DES in 2001.

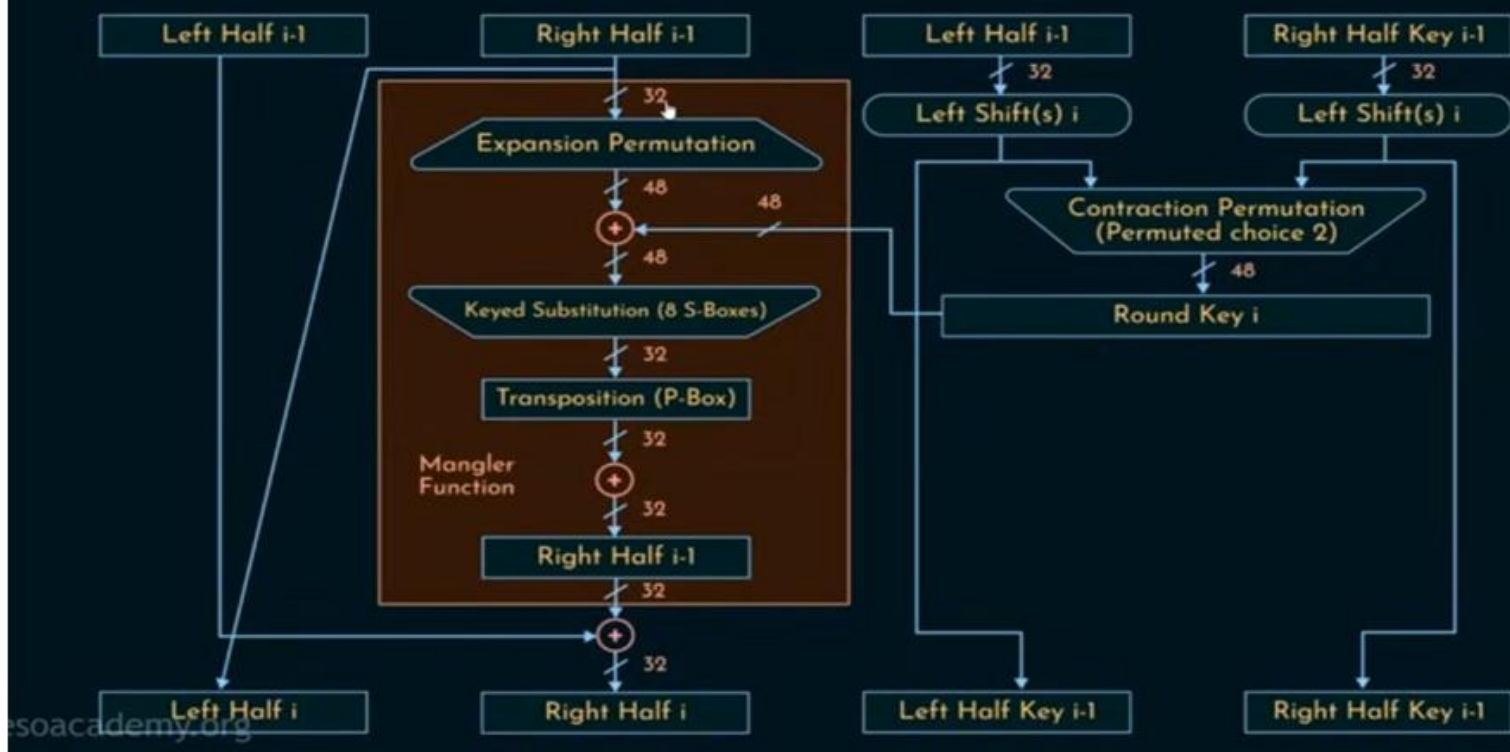
See on YouTube how this works to learn algorithms and solve numericals faster.

- **Advanced Encryption Standard (AES)**
- **Data Encryption Standard (DES) works.**
- **Study working examples of Inverse SubBytes in AES.**
- **practice AES Inverse MixColumns with step-by-step calculations.**

The Symmetric Cipher Model:



Single Round of DES Algorithm

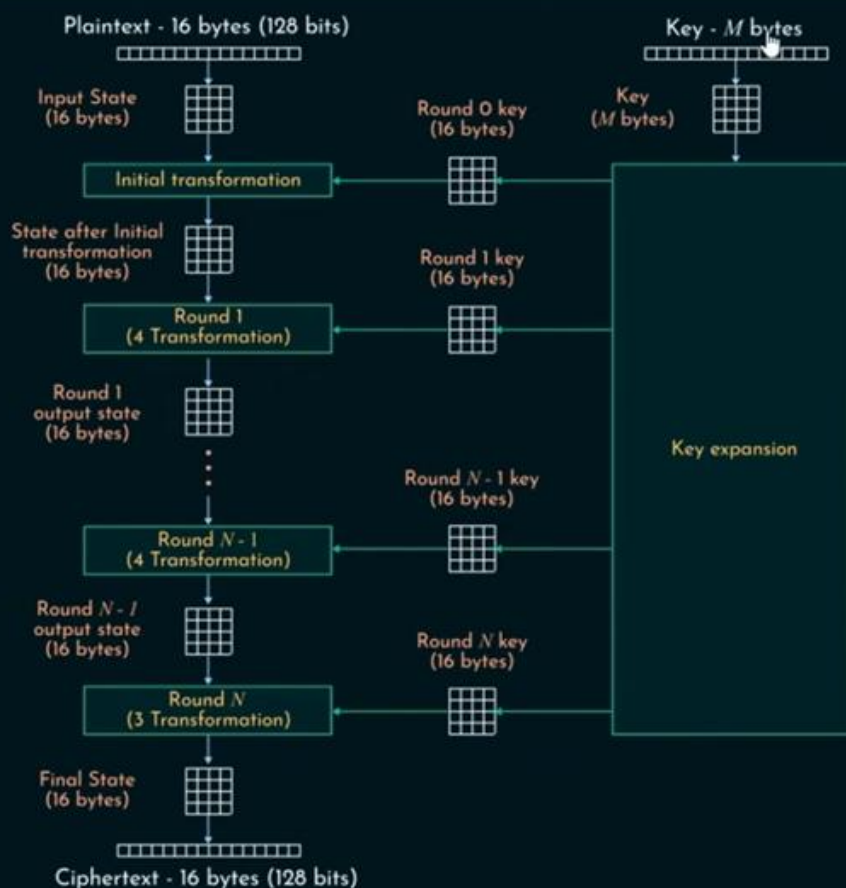


DES Encryption Algorithm



AES Structure

No. of rounds	Key size (in bits)
10	128
12	192
14	256



AES Parameters

	AES-128	AES-192	AES-256
Key Size	128	192	256
Plaintext Size	128	128	128
Number of rounds	10	12	14
Round Key Size	128	128	128

Creation of Round Keys

